

An Empirical Comparison of Option-Pricing Models in Hedging Exotic Options

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This paper examines the empirical performance of various option-pricing models in hedging exotic options, such as barrier options and compound options. A practical and relevant testing approach is adopted to capture the essence of model risk in option pricing and hedging. Our results indicate that the exotic feature of the option under consideration has a great impact on the relative performance of different option-pricing models. In addition, for any given model, the more “exotic” the option, the poorer the hedging effectiveness.

Since the publication of the path-breaking contribution by Black and Scholes (1973), more realistic and complicated models have been proposed in the option-pricing literature. For example, deterministic volatility function models assume that the volatility of the underlying asset depends on both the price of the underlying asset and time (Derman and Kani, 1994). Another example is the jump diffusion models, which allow the stochastic process of the underlying asset to have discontinuous breaks (Merton, 1976). Stochastic volatility models, on the other hand, assume that the volatility follows a particular stochastic process (Heston, 1993). While these alternative models provide important theoretical insights, they are motivated primarily by their analytical tractability. Moreover, each model differs fundamentally in its implications for valuing and hedging derivative contracts. Consequently, a test of the empirical validity and performance of these models is necessary before they can be applied fully in practice. It is also important to determine the most suitable model when each particular option is considered.

The performance of various option-pricing models has been studied extensively in the existing literature. Bakshi, Cao, and Chen (1997) explore the pricing and hedging performance of a comprehensive model that includes the Black and Scholes (1973) (BS) model, the stochastic volatility and stochastic interest rate model, and the stochastic volatility and jump diffusion model as special cases. Their results indicate that the alternative models outperform the BS model in terms of out-of-sample pricing errors. The hedging performance, however, is relatively insensitive to model misspecifications. Dumas, Fleming, and Whaley (1998) evaluate a few deterministic volatility function models and demonstrate that they perform no better in out-of-sample pricing and hedging than the implied volatility model. The testing approach in most of

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the existing empirical work in current literature is common. The parameters of the model under consideration are estimated such that the model prices for some European options match those prices that are observed in the market (e.g., from market transactions or broker quotes) at a specific time. The resulting model is used to price other European or American options later. Next, these model prices are compared to the prices observed from the market at this time. However, this out-of-sample test does not fully capture the essence of model risk, as option-pricing models are chiefly used to price or hedge exotic or illiquid options at the time they are calibrated. Moreover, model specification is relatively less important when vanilla options are concerned, since they are actively traded in the market and a great deal of information on the value of these options is readily available.

The purpose of this paper is to evaluate the BS model and three other major alternative models: 1) the jump diffusion (JD) model, 2) the stochastic volatility (SV) model, and 3) the stochastic volatility and jump diffusion (SVJ) model using a different, but more appropriate approach. The methodology employed in this paper is practical and relevant, as it properly addresses the issue of model risk in option pricing and hedging. First, the models under consideration are calibrated to the observed cross-sectional vanilla option prices and the resulting models are used to set up replicating portfolios for other options at the same time. The models are frequently recalibrated to ensure that they stay close to the market when used. This is a cross-sectional test rather than a time series out-of-sample test and is consistent with how practitioners use the models.

Second, the test is based upon the performance of the models in hedging exotic options, such as barrier options and compound options.¹ It is more appropriate to assess model risk based on exotic options rather than liquid or European options. This is because exotic options have more complex payoff structures than their vanilla counterparts. As such, they are more sensitive to model misspecifications and could be seriously mispriced or mishedged by a model that might otherwise accurately value European options. Furthermore, exotic options are traded in the over-the-counter market; thus, historical data are not readily available. As a result, practitioners rely heavily on models to value these contracts and model risk becomes particularly important where they are concerned.

Intuition indicates that the performance of any given model in pricing or hedging exotic options should depend on how important the exotic feature of the option is. Consider, for example, barrier and compound options. Both types of options are path dependent in the sense that their payoffs depend on the underlying asset price at more than one time. However, the payoff of a barrier option is much more dependent on the joint distribution of the underlying asset price at different points in time than is the payoff for a compound option. In other words, barrier options are further away from European options than compound options in terms of their payoff structures. As a result, we expect that the hedging of a barrier option will be more risky than that of a compound option. Moreover, for different barrier options, the impact of the barrier feature on their payoffs may also differ significantly. For instance, the barrier feature is much more important for an option with a barrier closer to its underlying asset price than it is for other options as there is a greater probability that the barrier will be reached during the option's life. Therefore, the sensitivities to the model specification are different for options with different barrier levels. For this reason, it is crucial to investigate whether and how the exotic feature in options affects the model performance.

In order to overcome the lack of historical data on exotic options, we examine the accuracy of hedging a synthetically created exotic option. The model parameters are estimated at the current

¹As an example, we consider up-and-out call options and call-on-call options in this paper.

time from market prices of traded assets and a replicating portfolio is established for a particular exotic option based on the model under consideration. At the next point in time, the model is recalibrated, the model price for the exotic option is compared with the value of the replicating portfolio, and their difference is then defined as the hedging error over the rebalancing interval. The replicating portfolio is then rebalanced with the new hedging parameters. This procedure continues until the exotic option matures and the average hedging errors are used as an indicator of the model's hedging effectiveness.

Our testing methodology is similar to that of Melino and Turnbull (1995), who examine the effect of stochastic volatility on the pricing and hedging of long-term foreign currency options. Since long-term foreign currency options are not actively traded and there are no data available, they look at how much the value of the replicating portfolio deviates from the model price. However, our method is different in that our model is recalibrated frequently and the test is based on exotic options, which are not among the securities whose prices are used in the calibration process. Green and Figlewski (1999) investigate the performance of the BS model when recalibrated daily to historical data. In contrast, we recalibrate the model to current market prices of traded options. Hull and Suo (2002) adopt a similar approach to ours; however, instead of using the market data, they assume there is a "true" model and the vanilla and exotic option prices are generated from it. They also compare the pricing errors from a calibrated deterministic volatility function model.

Fink (2003) presents a static hedging algorithm and conducts an analysis of the effectiveness of hedging barrier options when volatility is stochastic. Nalholm and Poulsen (2006) examine the static hedging and model risk for barrier options. By looking at barrier options, these studies address the essence of model risk in options but are limited in their focus on static hedging where the replicating portfolios are not rebalanced until the maturity of the target option once they have been constructed. Therefore, they cannot reflect the changes in market conditions during the life of the option being hedged. In our dynamic hedging procedure, the model is frequently recalibrated and the hedge is rebalanced at each liquidation date. Consequently, in this paper, we essentially evaluate a practitioner's version of option-pricing models.

Our findings demonstrate that the exotic feature of the hedged option has a great impact on the relative performance of different models. The SV model outperforms the BS model in hedging all types of the up-and-out call options under consideration except the long-term out-of-the-money options. The jump feature in the JD and SVJ models cannot effectively capture the stylized facts of excess kurtosis and stochastic skewness observed in the currency options market when these models are used to hedge barrier options. Both models generally produce larger hedging errors than the BS model, especially for hedging short-term and medium-term options. However, for hedging compound options, the alternative models usually perform better than the BS model, and the moneyness and maturity effects on the hedging effectiveness are the same as those in the case of hedging European options. This is because when compared with barrier options, the payoff structures of compound options are similar to those of vanilla options. Overall, we find that the inclusion of stochastic volatility provides the most improvement on hedging exotic options. For any given model, the hedging of long-term in-the-money up-and-out options is more risky than that of short-term otherwise the same type of options. The reason is that the probability of the underlying asset price reaching the barrier level for long-term options is higher than that for short-term options, and the knock out feature is more important for in-the-money options than for out-of-the-money options. In addition, the hedging of up-and-out options with lower barrier levels is more risky than the hedging of these options with higher barrier levels. This implies that the more "exotic" the option under consideration, the poorer the hedging performance.

The remainder of this paper is organized as follows. Section I briefly reviews various option-pricing models and examines the potential model risk for the corresponding practitioner's version of these models. Section II discusses the estimation and testing methodologies. Section III describes the data used in the analysis and presents the estimation results. In Section IV, we report the empirical results, while Section V concludes this paper.

I. Option-Pricing Models

A. Theoretical Models

1. The BS Model

The BS model assumes that the underlying asset price S follows a geometric Brownian motion under the risk-neutral probability measure

$$dS/S = (r - q) dt + \sigma dw, \quad (1)$$

where r is the risk-free rate, q is the dividend yield, σ is the volatility of the underlying asset and is a constant in this model, and w is a standard Brownian motion. For simplicity, r and q are assumed to be constant throughout this paper. For currency options, q is equal to the foreign interest rate.

European options can be valued analytically under the BS model. For a European call option with strike X , its price $C(t, S, X, \sigma)$ at time t is given by

$$C(t, S, X, \sigma) = Se^{-q(T-t)}N(d_1) - Xe^{-r(T-t)}N(d_2), \quad (2)$$

where

$$d_1 = \frac{\ln(S/X) + (r - q + 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t}.$$

Under the assumptions of the BS model, implied volatilities should be the same for options on the same asset with different strikes. However, it is recognized that the implied BS volatilities vary systematically with strikes and maturities. In the currency options market, the implied volatilities for away-from-the-money options with the same maturity are usually higher than those of at-the-money options. This volatility pattern, called the volatility smile, indicates that the implied currency return distribution is positively/negatively skewed with higher kurtosis than allowable in the lognormal distribution assumed by the BS model. To capture these stylized facts observed in empirical studies, researchers proposed a number of alternative models that relax the assumptions on either the volatility or the continuous dynamic process of the underlying asset price.

2. The JD Model

Merton (1976) proposes a jump diffusion process to model the stock price subject to occasional discontinuous breaks. The model assumes that under the risk-neutral probability measure, the underlying asset price follows:

$$dS/S = (r - q - \lambda\mu) dt + \sigma dw_t + JdQ, \quad (3)$$

where:

- σ is the volatility of the underlying asset returns (conditional upon no jump occurring), λ is the annual frequency of jumps, μ is the average jump size measured as a proportional increase in the asset price;
- J is the random percentage jump conditional on a jump occurring, and

$$\ln(1 + J) \sim N(\ln(1 + \mu) - 0.5\delta^2, \delta^2);$$

- Q is a Poisson counter with intensity λ , i.e., $\text{Prob}(dQ = 1) = \lambda dt$, and dQ is assumed to be independent of dw .

In this model, the instantaneous mean of the underlying asset returns consists of two parts: 1) the first part is due to the normal underlying asset price changes, and 2) the second part is due to the abnormal underlying asset price changes. Accordingly, the variance of the total return of the underlying asset has two components as well: 1) the component of the normal time variance, and 2) the component of jump variance. If there is no jump, $\lambda = 0$, then this model reduces to the BS model. Compared to the BS model, the jump diffusion model attributes the skewness and excess kurtosis observed in the implied distribution of the underlying asset returns to the random jumps in the underlying asset returns. The skewness arises from the average jump size and the excess kurtosis arises from the magnitude and variability of the jump component. Therefore, the jump diffusion model could be more capable of capturing the empirical features of underlying asset returns than the BS model.

For a European call option, its price $C(t, S, X, \lambda, \mu, \delta)$ is given by (Merton, 1976)

$$C(t, S, X, \lambda, \mu, \delta) = e^{-r(T-t)} \sum_{n=0}^{\infty} \frac{e^{-\lambda(T-t)} (\lambda(T-t))^n}{n!} (S_t e^{r_n(T-t)} N(d_{1n}) - X N(d_{2n})), \quad (4)$$

where

$$r_n = r - q - \lambda\mu + \frac{n \ln(1 + \mu)}{T - t},$$

$$d_{1n} = \frac{\ln(S_t/X) + (r_n(T-t) + 0.5(\sigma^2(T-t) + n\delta^2))}{\sqrt{\sigma^2(T-t) + n\delta^2}}, \quad d_{2n} = d_{1n} - \sqrt{\sigma^2(T-t) + n\delta^2}.$$

3. The SVJ Model

Bakshi, Cao, and Chen (1997) develop a model in which the underlying asset price is allowed to have discontinuous breaks and the return variance is assumed to follow a mean-reverting square root process. Namely, under the risk-neutral probability measure, the underlying asset return and volatility processes are as follows:

$$\begin{aligned} dS/S &= (r - q - \lambda\mu) dt + \sqrt{v} dw + JdQ \\ dv &= k(\theta - v) dt + \sigma\sqrt{v} dz, \end{aligned} \quad (5)$$

where:

- v is the instantaneous variance conditional on no jump occurring;
- w and z are standard Brownian motions with a correlation coefficient ρ ;
- k , θ , σ are the speed of adjustment, long-run mean, and volatility of volatility parameters, respectively;
- the parameters of the jump component are the same as those described in the JD model.

If there is no jump, $\lambda = 0$, then this model reduces to Heston's (1993) SV model. If the volatility is constant, this model parallels the jump diffusion process specified in Equation (3). Such a model provides some additional flexibility over the BS model and the JD model in capturing the empirical features found in the distribution of the underlying asset returns. It attributes the skew effect to the correlation between the underlying asset returns and the volatility, the volatility of volatility, or the average jump size, and it attributes the kurtosis effect to the volatility of volatility or the magnitude and variability of the jump component. Another advantage of this framework is that European options can still be valued analytically.

For a European call option written on the asset with strike price X and maturity T , Bakshi, Cao, and Chen (1997) show that its price $C(t, S, v, X)$ is given as

$$C(t, S, v, X) = S e^{-q(T-t)} P_1 - X e^{-r(T-t)} P_2, \quad (6)$$

where

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-i\phi \ln(X)} f_j(t, T, S, v; \phi)}{i\phi} \right] d\phi$$

for $j = 1, 2$. f_j are the characteristic functions of P_j , respectively, and are given in the Appendix.

This paper considers the Bakshi, Cao, and Chen (1997) SVJ model, as well as its special case, the SV model.

B. Practitioner's Model

Although the alternative models are more realistic, the most widely used valuation procedure among practitioners is still the BS model, with ad hoc adjustments and recalibrations. This so-called practitioner's Black-Scholes (PBS hereafter) approach can be described as follows. The BS implied volatilities of all traded options are calculated at the current time, and a volatility surface is built by interpolation across strike prices and maturities. With this volatility surface, other option prices are calculated from the BS formula using the volatility obtained from the corresponding point on the surface. This procedure repeats whenever the model is used.

The key point of the PBS is to calibrate and recalibrate the BS model to fit the European option prices. In other words, market participants tend to calibrate their models to fit the observed cross-sectional market prices at a particular time, and then recalibrate them whenever they mark the option to market or rebalance their hedging portfolios. By doing this, the model stays close to the market and the parameters are allowed to change over time, which may add flexibility to each model to capture changes in the distribution of the underlying asset returns that models with constant parameters fail to capture.

Moreover, market participants mainly use models to price or hedge illiquid or exotic options. In fact, the practitioner's model uses all observed liquid option prices as the inputs to the model

instead of the outputs from the model. Other models are also used by practitioners in a similar fashion.

C. Model Risk in the Practitioner's Model

The practitioner's approach can fit the market-traded options closely when the model is recalibrated; however, it does not eliminate the model risk. First, the dynamics of the underlying asset price obtained by fitting the model to a cross-section of observed option prices may be incompatible with the no arbitrage evolution of the underlying asset price. Second, periodically updating the model implicitly assumes that the fitted parameters can change over time implying that the model is internally inconsistent and it may permit arbitrage opportunities in derivatives (Backus, Foresi, and Zin, 1998).

The model risk becomes especially important when the model is used to price or hedge exotic options. This is because fitting all European option prices means the model will give the correct unconditional probability distribution of the underlying asset price at any particular future time. However, different models will give different joint distribution of the underlying asset price at different points in time. Consequently, even if the practitioner's version of the model can correctly price a derivative whose payoff depends on the asset price at any one particular time, there is no guarantee that it can correctly price a derivative whose payoff depends on the underlying asset price at more than one time (such as the case with barrier options and compound options). In addition, most exotic options are traded in the over-the-counter market and their market prices are not readily available. Practitioners cannot calibrate and recalibrate their models to the market prices of exotic options in the same way they can with vanilla options. As a result, exotic option prices are more sensitive to the model misspecification than vanilla option prices. For these reasons, it makes more sense to look at exotic options from the practitioner's point of view when assessing the model risk.

Frequent recalibration of a model might also generate hedging errors. To illustrate this point, we take the BS model as an example. Assume that the option is an exotic option whose price is not available, and one has to rely on the model for pricing and hedging. In the BS model, the market is complete, so the exotic option can be perfectly hedged by taking the underlying asset and the risk free investment as hedging instruments. Since the replicating portfolio can only be rebalanced discretely in practice, hedging errors may arise from both discrete adjustments to the hedging portfolio and model misspecifications.

Assume that the true volatility of the underlying asset price is σ_t , and it is misspecified as $\bar{\sigma}_t$ in the BS model at time t .² Denote the true price and the model price of an option by $C(t, \sigma_t)$ and $\bar{C}(t, \bar{\sigma}_t)$, respectively.³ The replicating portfolio based on the misspecified model consists of $\Delta_t = \partial \bar{C} / \partial S$ units of S_t and a risk free asset B_t at time t . The value of the portfolio is

$$\pi_t = -\bar{C}(t, \bar{\sigma}_t) + \Delta_t S_t + B_t. \quad (7)$$

It is self-financing if $\pi_t = 0$, which implies that $B_t = \bar{C}(t, \bar{\sigma}_t) - \Delta_t S_t$. The option price in Equation (7) is the model price $\bar{C}(t, \bar{\sigma}_t)$ because the market price for this option is not available. Nevertheless, such a hedging strategy is still useful in identifying models that can set up more

²For simplicity, the dividend rate q is assumed to be zero in the following derivation of the hedging errors.

³We assume that the option price depends on underlying asset price only through current price. This may not be true for path-dependent options.

accurate hedges for the target option. As the model is recalibrated at each rebalancing time, the hedging error from time t to time $t + dt$ is given by

$$\begin{aligned} d\pi_t &= \pi_{t+dt} - \pi_t \\ &= -[\bar{C}(t + dt, \bar{\sigma}_{t+dt}) - \bar{C}(t, \bar{\sigma}_t)] + \Delta_t dS_t + B_t r dt \\ &= -[\bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t, \bar{\sigma}_t)] + [\bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t + dt, \bar{\sigma}_{t+dt})] \\ &\quad + \Delta_t dS_t + B_t r dt. \end{aligned} \quad (8)$$

If the high-order terms of $d\bar{C}(t, \bar{\sigma}_t)$ are ignored, then the following is obtained

$$d\bar{C}(t, \bar{\sigma}_t) = \bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t, \bar{\sigma}_t) = \frac{\partial \bar{C}}{\partial t} dt + \frac{\partial \bar{C}}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 \bar{C}}{\partial S^2} S_t^2 \sigma_t^2 \varepsilon^2 dt, \quad (9)$$

where ε is a random number drawn from the standard normal distribution. Note that $\bar{C}(t, \bar{\sigma}_t)$ satisfies the following partial differential equation

$$\frac{\partial \bar{C}}{\partial t} + \frac{\partial \bar{C}}{\partial S} r S + \frac{1}{2} \frac{\partial^2 \bar{C}}{\partial S^2} S_t^2 \sigma_t^2 = r \bar{C}(t, \bar{\sigma}_t).$$

Plugging Equation (9) into Equation (8) yields

$$\begin{aligned} d\pi_t &= -\left(\frac{\partial \bar{C}}{\partial t} dt + \frac{\partial \bar{C}}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 \bar{C}}{\partial S^2} S_t^2 \sigma_t^2 \varepsilon^2 dt\right) \\ &\quad + [\bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t + dt, \bar{\sigma}_{t+dt})] + \Delta_t dS_t + B_t r dt \\ &= -\left(\frac{\partial \bar{C}}{\partial t} dt + \frac{\partial \bar{C}}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 \bar{C}}{\partial S^2} S_t^2 \sigma_t^2 \varepsilon^2 dt\right) \\ &\quad + [\bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t + dt, \bar{\sigma}_{t+dt})] + \frac{\partial \bar{C}}{\partial S} dS_t + \left(\bar{C}(t, \bar{\sigma}_t) - \frac{\partial \bar{C}}{\partial S_t} S_t\right) r dt \\ &= \frac{1}{2} \frac{\partial^2 \bar{C}}{\partial S_t^2} S_t^2 (\bar{\sigma}_t^2 - \sigma_t^2 \varepsilon^2) dt + [\bar{C}(t + dt, \bar{\sigma}_t) - \bar{C}(t + dt, \bar{\sigma}_{t+dt})]. \end{aligned} \quad (10)$$

The first term on the left-hand side of Equation (10) arises from both the model misspecification as well as the discrete adjustments to the hedge. The expectation and variance of this term are zero if the model is correctly specified and the hedging portfolio is rebalanced continuously. Furthermore, the size of this term is proportional to the model's gamma-hedge parameter. As a result, exotic options and vanilla options may have different sensitivities to model misspecification. Intuitively, this is because the payoff of an exotic option depends not only on the underlying asset price at maturity, but also on its price throughout the life of the option. The second term in Equation (10) arises from the change of model prices at time $t + dt$ due to the model recalibration. If the model is correctly specified and the hedging portfolio is frequently rebalanced, Galai (1983) documents that the discrete adjustment errors are relatively small compared with the misspecification hedging errors.

II. Empirical Methodologies

A. Parameter Estimation

There are usually two different ways to estimate the parameters of a given model. One is to use econometric methods (e.g., maximum likelihood estimation) with historical data of the underlying asset prices. One of the potential problems of this approach, as noted by Bakshi, Cao, and Chen (1997), is its stringent reliance upon historical data. In addition, it is impossible to obtain the parameter estimates for the risk-adjusted processes for some models that are necessary for the purpose of option valuation (e.g., the stochastic volatility model). The other way is to imply the parameters from the observed option prices. Various empirical studies have shown that implied parameters are better estimators as they reflect the market participants' view of the future movements of the asset price (Bates, 1996).

For these reasons, we use the latter method in model estimation. For a model that depends on a set of parameters $\Theta = (a_1, a_2, \dots, a_n)$, let us write the price of a vanilla option (call or put) with strike X and maturity time T as $C(t, \Theta, S, X, T)$, where t and S represent the current time and underlying asset price, respectively. At each time t , there are usually many vanilla options with different strikes and maturities traded in the market place. If we denote the corresponding market prices as $\tilde{C}(t, S, X, T)$, then the parameter vector Θ at time t is chosen to minimize the sum of the squared percent errors (SSE%)

$$\text{SSE\%}(t) = \min_{\Theta} \sum_{i,j} \left(\frac{\tilde{C}(t, S, X_i, T_j) - C(t, \Theta, S, X_i, T_j)}{\tilde{C}(t, S, X_i, T_j)} \right)^2. \quad (11)$$

This estimation process repeats whenever the model is needed to either price or hedge options.

Different objective functions used in the model estimation may yield different estimation and performance results (Christoffersen and Jacobs, 2004). The percent error function has been chosen in this paper because it is commonly used in existing literature and leads to a better weighting scheme as compared to other functions, such as the sum of squared errors (Detlefsen and Härdle, 2007).

B. Hedging Effectiveness

The hedging effectiveness is measured from the perspective of traders who want to minimize the uncertainty of their derivative positions via dynamic hedging. To parallel the standard market practice of applying option-pricing models, we frequently recalibrate the models and focus on exotic options as the target options in the test procedure. As exotic options are traded in the over-the-counter markets and historical data are unavailable, we look at the errors between the price predicted by the model and the value of the replicating portfolio.

Specifically, we adopt the following approach for testing the effectiveness of a particular model in valuing or hedging an exotic option

1. We first divide the interval $[0, T]$ into m equal steps and assume that hedging portfolios are rebalanced only at these times. Parameters of the model are estimated using the cross-sectional vanilla option prices observed from the market at each time $j\Delta t$. The resulting model is used to price the exotic option and to set up the replicating portfolio.
2. At the next time $(j+1)\Delta t$, model parameters are reestimated. The model price of the exotic option is compared with the value of the replicating portfolio that has been set up at

- $j \Delta t$ with the difference denoted by $\Delta\pi_j$. The replicating portfolio is rebalanced based on the new parameters estimated for the model at this time.
3. The above steps are repeated until the maturity of the target option. The hedging errors $\Delta\pi_j$, for $j = 1, 2, \dots, m$, are recorded.
 4. Consider a set of the same exotic options C_i ($i = 1, 2, \dots, n$), and repeat Steps 1 to 3. We have $n \times m$ hedging errors and denote the hedging error for the i th option at step j by $\Delta\pi_{ij}$.

Finally, we standardize the hedging errors by dividing them by the value of the hedged option at its inception in order to make them comparable across different options. Namely, the mean dollar hedging error and the mean absolute deviation (MAD) are calculated as

$$E(\Delta\pi) = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \frac{\Delta\pi_{ij}}{C_i}$$

$$\text{MAD}(\Delta\pi) = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \left| \frac{\Delta\pi_{ij}}{C_i} - E(\Delta\pi) \right|. \quad (12)$$

The average dollar errors measure the average percentage losses (or profits) of the hedging portfolios over the rebalancing interval, and the mean absolute deviation measures the variability of the errors over the rebalancing interval. Ignoring the errors from time discretization, the hedging errors should be zero if the model is specified correctly. Consequently, they can be used as indicators of the model's hedging effectiveness.

The exotic options considered in this paper include barrier options and compound options, which are among the most widely used options by market practitioners and academics alike. For barrier options, the barrier feature may have different impacts on the pricing and hedging of different types of options. For instance, when the underlying asset price is just below the barrier level, an up-and-out call is in-the-money and quite valuable; however, a small increase in the price may cause the option to be worthless. In contrast, a down-and-out call must be out-of-the-money just before it ceases to exist. Therefore, a slight error in estimating the knockout probability may have a larger impact on the pricing and hedging of up-and-out calls than it may have on those of down-and-out calls. We expect that the hedging of down-and-out calls is similar to that of vanilla options. For this reason, we focus on up-and-out calls in this paper. For compound options, we look at call-on-call options. However, with the exception of the BS model, there are no analytical formulas for the prices of these options. As a result, numerical methods have to be used to value these options and to calculate the hedging ratios.⁴

Some difficulties arise when hedging exotic options. Consider, for example, an up-and-out call option with strike price X and a barrier level of H . Its payoff at maturity T is $\max[S_T - X, 0]I_{\{\max_{0 \leq t \leq T} S_t < H\}}$, where $I_{\{A\}}$ is the indicator function of the event A . The delta and gamma of the option become large in absolute values near expiration when the asset price is close to the barrier. A trader who adopts the delta hedging strategy would take large short (or long) positions in the underlying asset and make large adjustments to the hedging portfolio (Engelmann, Fengler, Nalholm, and Schwendner, 2006). This will cause the hedging strategy to be very risky. To avoid

⁴We adopt the Monte Carlo simulation approach in this paper. The delta or vega ratio is the change in the simulated price of the exotic option for a 1% change in the underlying asset price or its volatility. We use 50 time steps and run 100,000 simulations for barrier options and 20,000 simulations for compound options. Broadie, Glasserman, and Kou's (1997) continuity correction for discrete barrier options is applied in the simulation.

such difficulties, the hedging positions are only rebalanced up to seven days before the maturity of the option.

III. Data Description and Model Parameter Estimation

A. Data Description

The data sets used for our analysis include the EUR/USD Currency Option Volatility Index and USD and Euro LIBOR rates for the period from January 2, 2002 to June 29, 2007, which are downloaded from the British Bankers' Association website. The foreign exchange market is the largest market in the world, and the use of currency derivatives to hedge against currency risk has increased dramatically in the past few decades (Bodnar, Hayt, Marston, and Smithson, 1995; Phillips, 1995; Howton and Perfect, 1998). The currency options market has some unique features that make it an interesting market for research. The currency option data contain daily quotes for the Black and Scholes (1973) implied volatilities (IV) for at-the-money (ATM) and out-of-the-money (OTM) European options with various maturities. The data set also contains the underlying Euro and USD exchange rate. The ATM implied volatility quotes have five maturities: 1) one week, 2) one month, 3) three months, 4) six months, and 5) one year.⁵ The OTM implied volatility quotes are available in the form of 25-delta risk reversals and 25-delta strangles with maturities of one month, three months, and one year. The USD and Euro LIBOR rates are used as the domestic and foreign interest rates to price the EUR/USD currency options. These rates have 14 maturities: one week, two weeks, and 1-12 months, and the rates for periods other than the above 14 maturities are calculated through linear interpolation. Overall, there are 1,371 observations in each data series, spanning 287 weeks with a cross-section of 11 implied volatility quotes per date. Figure 1 plots the time series of Euro and US dollar exchange rates and ATM implied volatilities for EUR/USD currency options with maturities of one month, three months, and one year, respectively.

In the over-the-counter currency options market, option prices are quoted in terms of the Black and Scholes (1973) volatilities, and moneyness in terms of delta, which is defined as the partial derivative of the Black and Scholes option price with respect to the underlying spot rate.⁶ Note that the delta of an ATM option is always 50% and the delta of a deep OTM option is close to zero, while the delta of a deep in-the-money (ITM) option is close to one. The ATM implied volatilities are measured by the volatilities for 50-delta options, whose strike is approximately equal to the underlying spot exchange rate. A 25-delta option is out of the money and its strike can be inferred from the Black and Scholes formula, given the implied volatility quote. A 25-delta risk reversal (RR25) includes a 25-delta long call option (C25) and a 25-delta short put option (P25), which implies that the risk reversal quote is the difference between the volatilities of the 25-delta call and put options. That is

$$RR25 = IV(C25) - IV(P25). \quad (13)$$

A 25-delta strangle (S25) is a combination of a 25-delta long put option and a 25-delta long call option; therefore, the strangle margin is the difference between the average of the volatilities of the included options and the ATM implied volatility

⁵The original data also contain the implied volatilities for ATM options with a maturity of two years. However, these volatilities are excluded, as our focus is primarily on options with a maturity of less than or equal to one year.

⁶The delta for a put option is negative, but the market practice is to quote the absolute value.

Figure 1. The Time Series of Euro and US Dollar Exchange Rates and at-the-Money (ATM) Implied Volatilities

The Euro and US dollar exchange rate is expressed in terms of the number of US dollars to pay for one Euro. The implied volatility quotes are those for ATM EUR/USD currency options with maturities of one month, three months, and one year, respectively. The daily data are from January 2, 2002 to June 29, 2007, 1,371 observations for each series.

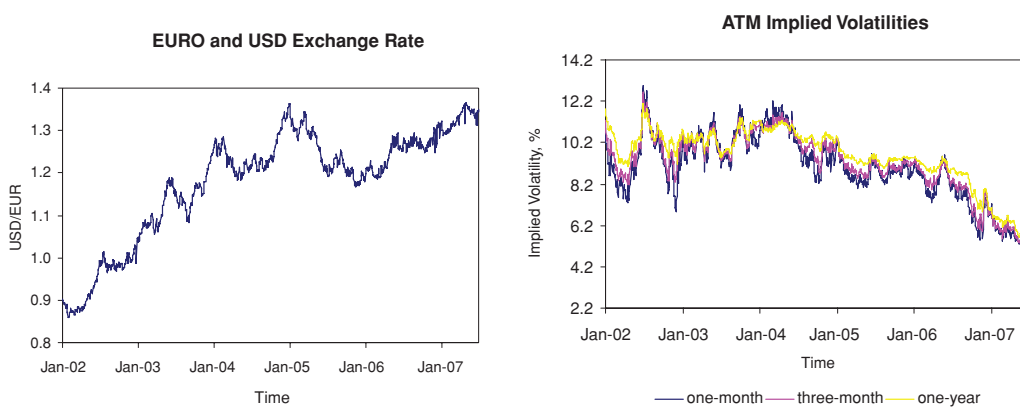
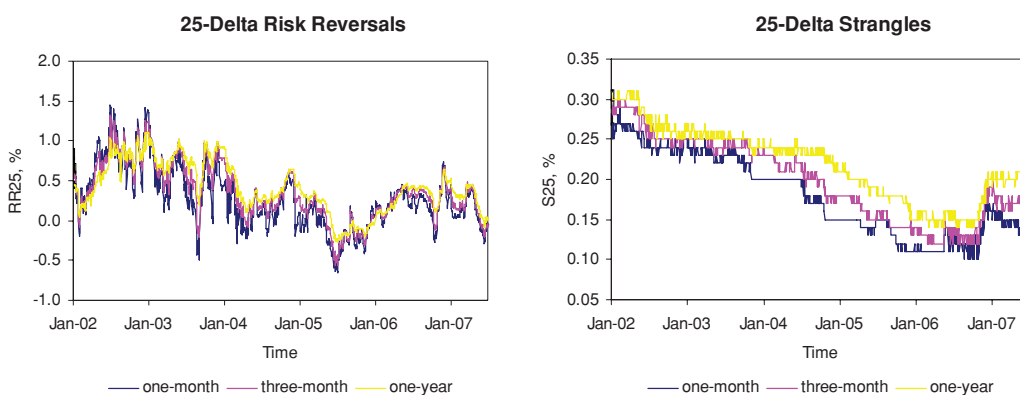


Figure 2. The Time Series of Risk Reversals and Strangles

The daily data are from January 2, 2002 to June 29, 2007, 1,371 observations for each series. The 25-delta reversals and strangles have maturities of one month, three months, and one year, respectively.



$$S25 = (IV(C25) + IV(P25))/2 - IV(ATM). \quad (14)$$

Figure 2 plots the time series of 25-delta risk reversals and strangles in the sample period with maturities of one month, three months, and one year, respectively. The figure illustrates that the 25-delta risk reversal quotes vary greatly over time regardless of the maturity, while the 25-delta

strangles are relatively stable. Note that the risk reversal measures the asymmetry of the implied volatility smile, and the strangle measures its curvature. This implies that the skewness of the risk-neutral currency return distribution is stochastic, which is consistent with the observation in the literature (Carr and Wu, 2007).

Given the market quotes for the 25-delta risk reversals and strangles and those for the ATM options, the implied volatilities for 25-delta call and put options can be solved from Equations (13) and (14). Following Carr and Wu (2007), we approximately denote the 25-delta call as a 75-delta put and the 25-delta put as a 75-delta call. Table I reports the implied Black and Scholes (1973) volatilities for 75-delta, ATM, and 25-delta call options with maturities of one month, three months, and one year. The implied volatilities are obtained by averaging the individual Black and Scholes implied volatilities within each moneyness maturity category and across the days in the sample. We can see that the average implied volatilities for the OTM and ITM options are indeed higher than those for the ATM options, and they are slightly asymmetric about ATM. This volatility pattern persists as the options maturity increases.

Together with the EUR/USD spot rate and the LIBOR rates, the Black and Scholes (1973) implied volatilities are used to calculate the market prices for different currency options. In this calculation, the time-to-maturity of an option is measured by the number of calendar days between the valuation date and expiration date. As a result, 23,307 different option prices are obtained with 17 observations every day.

B. Model Parameter Estimation

Parameters for a given model are estimated by Equation (11) every week using the observed option prices. The summary statistics of the implied parameters for various models are reported in Table II.

For the BS model, the mean of the estimated volatility parameter is 0.09 over the sample period. The implied volatilities vary from week to week with a minimum value of 0.05 and a maximum value of 0.12. The stability of the estimated parameters can also be inferred from the coefficient of variation, which equals the ratio between the standard deviation and the mean. The implied values from the BS model indicate that they are rather stable with the coefficient of variation of 0.16.

The average of the estimated volatility conditional on no jumps in the JD model is 0.08, with the standard deviation of 0.02. This model attributes the skewness and excess kurtosis to the jump risk, where jumps occur with a mean annual frequency of 1.28 times and with a mean jump size of 0.06. Except for the volatility parameter, the coefficients of variation for other parameters are greater than 1.0, which implies that these estimates are quite unstable.

For the SV model, the average values of fitted spot volatility parameters $\sqrt{v}$, k , θ , and σ are 0.09, 1.28, 0.07, and 0.19, respectively. Except for the estimate of θ , their coefficients of variation are less than 0.5, meaning that they are estimated with high stability. The correlation coefficient between the underlying currency returns and its volatility changes is positive, with a mean of 0.19.

For the SVJ model, the average estimates of $\sqrt{v}$, k , θ , σ , and ρ are 0.08, 1.00, 0.32, 0.23, and 0.11, respectively. These implied parameters are different from those for the SV model because this model adds the random jump feature to the SV model, and skewness and excess kurtosis are generated partly by the stochastic volatility process and partly by the random jumps. The fitted model indicates that jumps occur with a mean annual frequency of 1.56 times per year and an average jump size of -0.01 .

The estimation results for the three alternative models indicate that the distribution of the underlying asset returns is fat-tailed and asymmetric, which are the features that the BS model

Table I. Summary Statistics of the EUR/USD Currency Option Volatilities

This table reports the average volatilities for EUR/USD currency call options within each moneyness-maturity category for the period from January 2, 2002 to June 29, 2007. The 25-delta and 75-delta call options are out-of-the-money and in-the-money options, respectively. The volatilities are expressed in percentages.

Maturity	Moneyness		
	25-delta	At-the-Money	75-delta
1 month	9.281	8.958	9.006
3 months	9.571	9.189	9.216
1 year	9.983	9.389	9.562

Table II. Implied Parameters and SSEs

This table reports the weekly averages of the estimated parameters for the Black and Scholes (1973) model (BS), the jump diffusion model (JD), the stochastic volatility model (SV), and the stochastic volatility and jump diffusion model (SVJ) using EUR/USD currency options data from January 2, 2002 to June 29, 2007. Standard deviations are in parentheses. SSE% stands for the weekly average of the sum of squared percent errors.

Models	Parameters								SSE%
BS	σ								0.447
	0.093 (0.02)								
JD	σ	λ	μ	δ					0.125
	0.084 (0.02)	1.276 (1.99)	0.055 (0.13)	0.192 (0.27)					
SV	k	θ	σ	ρ	$\sqrt{v}$				0.136
	1.280 (0.48)	0.067 (0.19)	0.186 (0.09)	0.190 (0.29)	0.091 (0.02)				
SVJ	k	θ	σ	ρ	λ	μ	δ	$\sqrt{v}$	0.133
	0.999 (0.52)	0.317 (0.93)	0.226 (0.10)	0.111 (0.35)	1.564 (1.89)	-0.006 (0.15)	0.027 (0.12)	0.083 (0.01)	

fails to capture. The improvement of the in-sample fit of the alternative models over the BS model is further evidenced by the SSE% of the models: the average of the SSE% for the BS model is 0.45, which is the highest among all of the models considered. The averages of SSE% for the JD model, the SV model, and the SVJ model are 0.13, 0.14, and 0.13, respectively. These alternative models give a much better in-sample fit than the BS model, which is expected as they have more parameters, and, therefore, allow for more degrees of freedom. However, if some of the parameters in the model are redundant and cause overfitting of the data, the model will be penalized with larger out-of-sample pricing errors and worse hedging performance.

Overall, the estimated models over the sample period reveal evidence of the parametric instability. Models with more parameters can improve the in-sample fitting performance over those with fewer parameters, but they yield less stable estimates based on the coefficients of variation. Note that theoretical models are derived under the assumption of constant parameters. The

divergence from theory indicates that these models fail to capture some features of the process of the underlying spot rate.

IV. Hedging Performance

To assess the hedging effectiveness of a model under consideration, we employ two different dynamic hedging strategies: 1) the minimum variance hedging strategy and 2) the delta-vega neutral hedging strategy.

A. Minimum Variance Hedging Strategy

The minimum variance hedging strategy involves only the underlying asset and the risk-free asset as the hedging instruments, and the hedging ratio, X_s , is determined by minimizing the variance of the hedging portfolio. Specifically, if an option trader writes one option with the value of C and relies on this strategy to hedge this position, the value of the hedging portfolio at time t is

$$H = -C + X_s S + B,$$

where $B = C - X_s S$ is the amount of risk free investment. The hedging portfolio is self-financing, and the change of H from t to $t + dt$ can be written as

$$dH = -dC + X_s dS + Br dt.$$

The total variance of dH is given by

$$\text{Var}(dH) = \text{Var}(dC) + X_s^2 \text{Var}(dS) - 2X_s \text{Cov}(dS, dC).$$

By minimizing $\text{Var}(dH)$, the hedging ratio can be solved as

$$X_s = \frac{\text{Cov}(dS, dC)}{\text{Var}(dS)}. \quad (15)$$

In the BS model, the market is complete and an option can be perfectly hedged, in theory, by taking positions in the underlying asset and the risk-free asset. In this case, the minimum variance hedging is the same as the delta-neutral hedging and the hedging ratio is the delta of the hedged option.

However, for the alternative models, the minimum variance hedging strategy is no longer perfect in the sense that one cannot perfectly replicate the payoff of an option by only taking positions in the underlying asset and the risk-free investment. For the general SVJ models, Bakshi, Cao, and Chen (1997) demonstrate that the minimum variance hedging ratio is given by

$$X_s = \frac{v}{v + V_j} \frac{\partial C}{\partial S} + \rho \sigma \frac{\partial C}{\partial v} \frac{v}{S(v + V_j)} + \frac{\lambda}{S(v + V_j)} (E[JC(t, S(1 + J))] - \mu C(t, S)), \quad (16)$$

where $V_j = \lambda \mu^2 + \lambda(e^{\delta^2} - 1)(1 + \mu)^2$.

In the case of the pure JD model, X_S can be similarly obtained as:

$$X_S = \frac{\sigma^2}{\sigma^2 + V_j} \frac{\partial C}{\partial S} + \frac{\lambda}{S(\sigma^2 + V_j)} (E[JC(t, S(1 + J))] - \mu C(t, S)). \quad (17)$$

It illustrates that if there is no jump risk ($\lambda = 0$), the minimum variance hedging is the same as the delta-neutral hedging. However, if there is jump risk, the impact will be reflected in the second term of Equation (17).

For the SV model, Equation (16) reduces to:

$$X_S = \frac{\partial C}{\partial S} + \frac{\rho \sigma}{S} \frac{\partial C}{\partial v}, \quad (18)$$

which indicates that if the volatility is deterministic or stock returns are uncorrelated with volatility changes, then the minimum variance hedging ratio is the same as the delta hedging ratio.⁷

The hedging procedure is described as follows. At time t , model parameters are estimated by fitting the model to the prices of the traded European currency options. The price of the exotic option under consideration, C_t , can then be calculated from the model. To hedge this exotic option, a replicating portfolio is constructed with X_S units of asset S_t , and B_t units of the risk-free asset, which is chosen so that the value of the hedging portfolio is zero. At time $t + \Delta t$, the hedging portfolio is rebalanced. Using model parameters estimated at time $t + \Delta t$, the value of the hedging portfolio is given by:

$$H_{t+\Delta t} = -C_{t+\Delta t} + X_S S_{t+\Delta t} + B_t(1 + r \Delta t).$$

Thus, $H_{t+\Delta t}$ is the hedging error over the rebalancing interval Δt . The above steps are repeated up to one week before the option's maturity date.

This procedure tracks the hedging errors for one realization of the hedged option. The procedure is reemployed in the sample period after it is completed and each represents a realization of a sample path. The average dollar errors and the MAD are calculated for each model through these hedging errors as described in Section II.

Table III reports the hedging errors when the target options are up-and-out call options. The barrier level of the option is set to be 1.1 times the underlying spot rate. To examine the moneyness and maturity effects, we consider options with different moneyness levels of 0.94, 0.97, 1.00, 1.03, and 1.06, respectively.⁸ We also consider three different maturities: 1) one month (short term), 2) three months (medium term), and 3) one year (long term).

Several observations are derived from the average hedging errors in Table III. Based on both the average dollar errors and the MADs, the SV model generally performs better than the BS model in hedging short-term, medium-term, and most of the long-term up-and-out call options. This suggests that incorporating the stochastic volatility into the model framework significantly improves the performance of hedging these types of barrier options, and it is in line with the general findings in recent literature when the target options are plain vanillas. However, the BS model performs no worse than the SV model in hedging long-term OTM options. The probability

⁷Poulsen, Schenk-Hoppé, and Ewald (2007) demonstrate that the minimum variance hedging ratio in stochastic volatility models should be computed by using the minimal martingale measure. However, their empirical results indicate that the issue is not of major practical importance. This is especially true in the currency options market, where the unconditional distribution of the currency returns is relatively symmetric. We thank the referee for pointing this out.

⁸The moneyness is defined as the ratio of the underlying spot rate to the strike price of the option.

Table III. Minimum Variance Hedging Errors for Barrier Options

This table reports the minimum variance hedging errors for each model and for each category of up-and-out call options. The barrier levels are set equal to 1.1 times the underlying spot rate. An option is a short-term option if its maturity is one month, a medium-term option if three months, and a long-term option if one year. Moneyness is determined by S/X , where S denotes the spot exchange rate and X is the exercise price. BS, JD, SV, and SVJ stand for the BS model, the jump diffusion model, the stochastic volatility model, and the stochastic volatility and jump diffusion model, respectively.

Maturity	Moneyness (S/X)	Dollar Errors (%)				Mean Absolute Deviations (%)			
		BS	JD	SV	SVJ	BS	JD	SV	SVJ
Short term	1.06	-0.62	-0.56	-0.56	-1.10	0.49	4.07	0.52	4.44
	1.03	-0.97	-1.24	-0.91	-1.81	1.62	6.82	1.46	7.66
	1.00	-1.75	-4.06	-1.95	-4.59	8.27	16.56	7.84	16.74
	0.97	-3.36	-10.73	-1.40	-6.14	30.49	39.76	24.59	29.54
	0.94	-6.06	-10.97	3.87	-8.83	70.61	63.99	42.02	35.46
Medium term	1.06	-0.32	-0.09	-0.06	-0.79	3.42	6.80	2.92	6.66
	1.03	-0.29	-0.22	0.05	-0.99	4.86	9.47	4.23	9.82
	1.00	-0.78	-0.99	0.13	-1.75	8.04	14.54	6.89	15.49
	0.97	-2.55	-4.33	-0.52	-3.17	14.06	25.29	12.61	25.43
	0.94	-8.48	-10.28	-0.99	-6.29	27.97	41.88	21.57	40.01
Long term	1.06	-0.92	0.70	0.25	-0.52	7.08	8.42	5.09	5.40
	1.03	-1.10	0.87	0.49	-0.54	8.23	9.65	6.09	6.50
	1.00	-1.23	0.98	1.02	-0.45	9.73	11.39	7.71	8.03
	0.97	-0.73	0.72	2.17	0.29	12.87	13.69	10.44	10.51
	0.94	-1.46	0.91	3.95	2.62	19.37	16.33	13.43	14.18

of the underlying spot rate hitting the barrier is small for the short-term and medium-term barrier options considered in our example, but it becomes relatively large when the maturity of the barrier option becomes longer. To illustrate this fact, we estimate the risk-neutral probabilities of the stock price reaching barrier levels for the target options with different maturities under the BS model.⁹ These probabilities are calculated for these options with parameters estimated at the starting date of each realization, and the averages for short-, medium-, and long-term options are 0.09%, 4.63%, and 30.18%, respectively. As a result, short-term barrier options are closer to vanilla options than the corresponding long-term barrier options. We may conclude that the relative model performance depends on the exotic feature of the hedged option. The overall performances of the JD and SVJ models are particularly poor for hedging short-term and medium-term options, indicating that adding random jumps to the model framework simply

⁹The real probabilities are different from the risk-neutral ones, as the drift term of the underlying asset price dynamics under the real probability measure is the instantaneous expected return rather than the difference between the domestic and foreign rates. The formula for calculating the risk-neutral probability of stock price reaching the barrier for the up-and-out option is given by

$$\text{prob} = N(h'_2) + (H/S)^{2(r-q-0.5\sigma^2)/\sigma^2} N(-h_2),$$

where

$$h'_2 = \frac{\ln(S/H) + (r - q - 0.5\sigma^2)T}{\sigma\sqrt{T}}, \quad h_2 = \frac{-\ln(S/H) + (r - q - 0.5\sigma^2)T}{\sigma\sqrt{T}}.$$

increases the model instability and, therefore, may generate larger hedging errors. As the maturity of the hedged option increases, it seems that the ability of both the stochastic volatility and random jump generating excess kurtosis and skewness in the distribution of currency returns increases. The nonzero average dollar errors (largely negative in our study) may reflect the market prices of nontraded risk factors, such as volatility risk and jump risk; however, the relationship between hedging errors and risk premia should be interpreted with caution. This is because the discrete adjustment to the hedge and the model error may yield misleading hedging results concerning the sign of the risk premia (Branger and Schlag, 2008).

For any given model, the hedging performance also depends on the importance of the exotic feature in the hedged option. Moving across moneyness at a fixed maturity, it is noted that ITM options are easier to hedge than OTM options in terms of both dollar errors and MADs. This is consistent with the observations when the hedged options are of European style (Jiang and Oomen, 2001). This is true regardless of the maturities of the hedged options and the models considered. Jiang and Oomen (2001) also document that hedging effectiveness improves with maturity of the hedged European options. Interestingly, the maturity effects depend on the moneyness in the case of hedging barrier options. For the ITM options, the longer the maturity, the larger the MADs; however, for the OTM options, the reverse is true. This is because long-term options are more likely to be knocked out than short-term options for the given barrier level, and the barrier feature is more important for ITM options than for OTM options. This confirms Hull and Suo's (2002) conclusion that model performance depends on the degree of path dependence of the exotic option considered. For any given model, the more severe the exotic features of the option being hedged, the poorer the model is expected to perform.

It is expected that barrier levels of the hedged options have a great impact on the model's hedging effectiveness, as options with a lower knockout level are obviously more "exotic" than those with higher levels. To investigate this effect, we focus on the BS model and the SV model and further look at a few ITM and ATM up-and-out call options with the barriers set at 1.05 (lowest), 1.10 (lower), 1.20 (higher), and 1.30 (highest) times the underlying spot rate, respectively. As we can see from the results in Table IV, both models perform the worst for hedging options with the lowest barrier level in terms of MADs, and their performance improves as the barrier level of the option becomes higher. For ITM options, hedging performance also deteriorates as the maturities increase for any given barrier level. It is also interesting to note that the hedging errors for short-term and medium-term options with the higher barrier level are similar to those for the corresponding options with the highest barrier level. The reason for this is that when the barrier level is sufficiently high and the maturity is not too long, the up-and-out option is essentially a plain vanilla option and the barrier level has little impact on hedging performance.

Table V reports the hedging errors for the call-on-call options. The underlying call option is a one-month EUR/USD currency option with the strike equal to the underlying spot EUR/USD exchange rate. Strikes of the compound options are set at 0.5, 1.0, 1.5, 2.0, and 2.5 times the one-month ATM option price, respectively.

Based on the dollar hedging errors, the JD model performs slightly worse than the BS model, and the SV and SVJ models perform much better than the BS model. In this case, incorporating stochastic volatility into the model significantly improves the hedging effectiveness and the SVJ generally performs the best. This is much more pronounced for medium-term and long-term options. This can be explained by the fact that these options have a relatively low degree of path dependence on the underlying spot rate. The MADs, however, suggest that the effectiveness of

Table IV. Hedging Errors for Barrier Options with Different Barrier Levels

This table reports both the minimum variance and delta-vega hedging errors for the BS and SV models and for up-and-out call options with different barrier levels. The options are in-the-money (ITM, $S/X = 0.94$) and at-the-money (ATM, $S/X = 1.00$), and the barrier levels are set equal to 1.05 (lowest), 1.10 (lower), 1.20 (higher), and 1.30 (highest) times the underlying spot rate, respectively. An option is a short-term option if its maturity is one month, a medium-term option if three months, and a long-term option if one year. Moneyness is determined by S/X , where S denotes the EUR/USD spot rate and X is the exercise price. BS and SV stand for the BS model and the stochastic volatility model, respectively.

Maturity	Barrier	Minimum Variance Hedging				Delta-Vega Hedging			
		Dollar Errors (%)		MADs (%)		Dollar Errors (%)		MADs (%)	
		BS	SV	BS	SV	BS	SV	BS	SV
Panel A. ITM Options									
Short term	Lowest	−0.33	−0.13	5.94	4.13	−1.45	−0.71	3.37	2.02
	Lower	−0.62	−0.56	0.49	0.52	−0.65	−0.61	0.44	0.46
	Higher	−0.61	−0.52	0.35	0.38	−0.61	−0.54	0.33	0.36
	Highest	−0.61	−0.52	0.35	0.38	−0.61	−0.54	0.33	0.36
Medium term	Lowest	−0.25	0.61	7.78	6.11	−1.69	−0.56	5.15	4.53
	Lower	−0.32	−0.06	3.42	2.92	−0.93	−0.66	2.10	1.84
	Higher	−0.57	−0.57	0.50	0.52	−0.58	0.62	0.35	0.39
	Highest	−0.61	−0.56	0.44	0.45	−0.61	−0.57	0.30	0.34
Long term	Lowest	−1.14	−0.29	12.11	5.76	−2.72	−1.66	9.10	6.95
	Lower	−0.92	0.25	7.08	5.09	−1.94	−0.57	5.74	6.71
	Higher	−0.99	−1.08	2.60	2.22	−1.19	−1.19	2.09	2.11
	Highest	−0.60	−0.63	0.86	0.81	−0.56	−0.57	0.71	0.88
Panel B. ATM Options									
Short term	Lowest	−1.09	−1.00	15.16	8.22	−2.43	−1.64	10.04	4.96
	Lower	−1.75	−1.95	8.27	7.84	−1.07	−1.11	3.21	3.40
	Higher	−1.72	−1.83	8.58	8.33	0.01	−0.88	2.97	3.28
	Highest	−1.72	−1.83	8.58	8.33	0.01	−0.88	2.97	3.28
Medium term	Lowest	0.01	2.16	15.56	11.10	−2.50	0.22	11.81	8.35
	Lower	−0.78	0.13	8.04	6.89	−1.34	−0.39	5.84	5.38
	Higher	−1.37	−1.17	4.08	3.87	−0.58	−0.35	2.47	2.88
	Highest	−1.49	−1.13	4.03	3.93	−0.67	−0.24	2.44	2.90
Long term	Lowest	−1.56	−0.28	14.42	5.78	−3.91	−1.65	11.48	8.05
	Lower	−1.23	1.02	9.73	7.71	−2.55	0.16	7.97	9.81
	Higher	−1.66	−1.69	3.96	3.30	−1.67	−1.45	3.33	3.48
	Highest	−0.89	−0.84	1.74	138	−0.57	−0.37	1.45	2.01

this hedging strategy is relatively insensitive to model specifications. In this case, the results are consistent with those of Bakshi, Cao, and Chen (1997).

In contrast with the findings in the case of barrier options, the average dollar errors and MADs for compound options suggest that the maturity and moneyness effects are similar to those documented by Jiang and Oomen (2001). Call-on-call options with a low strike are easier to hedge than those with a high strike. For any given model, the hedging errors generally decrease with the

Table V. Minimum Variance Hedging Errors for Compound Options

This table reports the minimum variance hedging errors for each model and for each category of call-on-call options. The underlying call option is a one-month at-the-money currency option. An option is a short-term option if its maturity is one month, a medium-term option if three months, and a long-term option if one year. BS, JD, SV, and SVJ stand for the BS model, the jump diffusion model, the stochastic volatility model, and the stochastic volatility and jump diffusion model, respectively. The strikes of the call-on-call options are set equal to 0.5, 1.0, 1.5, 2.0, and 2.5 times the one-month at-the-money option price, respectively.

Maturity	Strike	Dollar Errors (%)				Mean Absolute Deviations (%)			
		BS	JD	SV	SVJ	BS	JD	SV	SVJ
Short term	0.5	-1.54	-1.69	-1.61	-1.22	7.30	7.54	8.09	7.84
	1.0	-1.92	-2.27	-1.48	-1.14	10.85	10.94	10.98	12.63
	1.5	-2.26	-2.61	-1.22	-0.48	14.90	14.41	13.97	19.21
	2.0	-2.55	-2.54	-0.91	0.81	19.57	17.64	17.26	27.80
	2.5	-2.77	-1.95	-0.51	2.69	24.60	20.46	20.88	38.45
Medium term	0.5	-1.34	-1.60	-0.90	-0.27	3.88	4.41	5.38	4.29
	1.0	-1.59	-1.91	-0.99	-0.24	4.70	5.29	6.13	5.37
	1.5	-1.85	-2.28	-1.16	-0.21	5.55	6.24	6.95	6.54
	2.0	-2.13	-2.68	-1.39	-0.17	6.46	7.27	7.85	7.86
	2.5	-2.49	-3.15	-1.69	-0.11	7.49	8.38	8.84	9.40
Long term	0.5	-0.79	-0.88	-0.53	-0.40	1.69	2.01	2.24	1.59
	1.0	-0.82	-0.91	-0.57	-0.42	1.83	2.15	2.39	1.73
	1.5	-0.85	-0.93	-0.61	-0.43	1.98	2.29	2.54	1.87
	2.0	-0.88	-0.95	-0.64	-0.44	2.15	2.45	2.71	2.03
	2.5	-0.91	-0.97	-0.67	-0.44	2.34	2.63	2.88	2.19

maturity of the call-on-call options. The reason is that an increase in the maturity of a compound option does not affect the importance of its exotic feature, nor the degree of path dependence.

B. Delta-Vega Neutral Hedging Strategy

Under the alternative models, options can no longer be perfectly hedged by trading only the underlying asset and the risk-free asset. For example, options written on the same asset are required to hedge the additional volatility risk in the SV model. A portfolio is delta-vega neutral if the portfolio value is insensitive to the changes in the underlying asset price and its volatility.

Suppose we need to hedge one unit of short position in an exotic option $C(t, S_t)$ at time t . The replicating portfolio consists of a_t units of a European option $C^E(t, S_t)$, b_t units of the underlying asset S_t , and B_t units of the risk-free asset. The value of the portfolio at the time t is thus:

$$\pi_t = -\bar{C}(t, S_t, \Theta_t) + a_t C^E(t, S_t) + b_t S_t + B_t,$$

where Θ_t represents the set of model parameters at time t . This portfolio is self-financing and is delta-vega neutral if:

$$\pi_t = 0,$$

$$\frac{\partial \pi_t}{\partial S_t} = -\frac{\partial \bar{C}}{\partial S_t} + a_t \frac{\partial C^E}{\partial S_t} + b_t = 0,$$

$$\frac{\partial \pi_t}{\partial v_t} = -\frac{\partial \bar{C}}{\partial v_t} + a_t \frac{\partial C^E}{\partial v_t} = 0,$$

and the hedge parameters are:

$$a_t = \frac{\bar{C}_v}{\bar{C}_v^E}, b_t = \bar{C}_S - \frac{\bar{C}_v}{\bar{C}_v^E} C_S^E, B_t = \bar{C} - a_t C^E - b_t S_t, \quad (19)$$

where the subscripts denote the partial derivatives of $\bar{C}$ with respect to the variable. Based on the above strategy, the value of the hedging portfolio at time $t + \Delta t$ is:

$$\pi_{t+\Delta t} = -\bar{C}(t + \Delta t, S_{t+\Delta t}, \Theta_{t+\Delta t}) + a_t C^E(t + \Delta t, S_{t+\Delta t}) + b_t S_{t+\Delta t} + B_t(1 + r \Delta t).$$

This is also the delta-vega hedging error over the rebalancing interval, since the value of the hedging portfolio is zero at time t .

As in the case of the minimum variance hedging strategy, the position being hedged is a short position in an up-and-out call option, or a call-on-call option with various maturities and strikes. The options used as the hedging instruments in this strategy are ATM European call options with the same maturity as that of the hedged option.

To illustrate how the hedging errors are calculated under this strategy, let us consider hedging a one-month up-and-out call option. In this case, the hedging instrument is a one-month ATM European call option whose market price is available every day and the replicating portfolio includes a_t units of the European option, b_t units of the underlying asset, and a risk-free asset worth B_t . At a particular time, the fitted model is used to calculate these hedging ratios as shown in Equation (19). At the next rebalance time, the model is recalibrated, and the difference between the theoretical value for the barrier option and the market value of the replicating portfolio is calculated. This is the delta-vega error resulting from using the hedge ratios implied by the model over the rebalancing interval. The replicating portfolio is then reconstructed based on the information available at this time. In this reconstructed portfolio, the hedging instrument is a new one-month ATM call option.¹⁰ This procedure continues until one week before the maturity of the hedged option.

For both the BS model and the JD model, vega hedging is not necessary because volatility is a constant under the model assumptions. As pointed out by Bakshi, Cao, and Chen (1997), the inclusion of another option in the delta-vega neutral hedge may not only neutralize the volatility risk but also reduce the remaining gamma risk in the hedging portfolio. To give each model a fair chance in the comparison, we follow their approach in implementing the so called delta-vega neutral hedge for the BS model and the JD model.

The hedging results for up-and-out call options are reported in Table VI. Note that this strategy generally yields lower MADs than the minimum variance strategy, indicating that the results are

¹⁰The vega of a European option will get very close to zero if it is deep in (or out of) the money and this may result in unstable hedging results. To avoid this problem, ATM options with a fixed maturity are used when constructing the replicating portfolios at any rebalance time. We are grateful to the referee for suggesting this.

Table VI. Delta-Vega Hedging Errors for Barrier Options

This table reports the delta-vega hedging errors for each model and for each category of up-and-out call options. The barrier level is set equal to 1.1 times the spot rate. An option is a short-term option if its maturity is one month, a medium-term option if three months, and a long-term option if one year. Moneyness is determined by S/X , where S denotes the EUR/USD spot rate and X is the exercise price. BS, JD, SV, and SVJ stand for the BS model, the jump diffusion model, the stochastic volatility model, and the stochastic volatility and jump diffusion model, respectively.

Maturity	Moneyness (S/X)	Dollar Errors (%)				Mean Absolute Deviations (%)			
		BS	JD	SV	SVJ	BS	JD	SV	SVJ
Short term	1.06	-0.66	-0.72	-0.61	-0.96	0.44	0.62	0.46	0.64
	1.03	-0.83	-1.06	-0.89	-1.53	0.99	1.21	0.94	1.23
	1.00	-1.07	-1.51	-1.11	-4.25	3.21	3.70	3.40	5.64
	0.97	3.40	-4.39	-0.91	-4.45	16.29	20.61	12.46	17.32
	0.94	10.02	-10.69	-6.71	-8.54	43.33	48.99	23.09	15.48
Medium term	1.06	-0.93	-0.33	-0.66	-0.64	2.10	2.04	1.84	2.28
	1.03	-1.10	-0.27	-0.70	-0.69	3.21	3.21	2.83	3.50
	1.00	-1.34	0.02	-0.39	-0.55	5.84	6.02	5.38	6.73
	0.97	-2.32	-0.08	0.03	0.40	10.66	2.97	10.89	14.08
	0.94	-6.86	-1.05	1.15	1.85	22.16	26.92	18.13	26.00
Long term	1.06	-1.94	-0.28	-0.57	-0.35	5.74	5.75	6.71	6.12
	1.03	-2.27	-0.10	-0.36	0.81	6.71	7.15	7.96	7.55
	1.00	-2.55	0.37	0.16	1.76	7.97	9.32	9.81	9.93
	0.97	-2.23	1.53	1.20	3.86	10.86	12.72	12.66	14.39
	0.94	-0.34	3.75	2.57	7.44	17.05	16.57	14.95	21.27

less variable under this strategy. It is recognized that the delta-vega hedging effectiveness greatly depends on the hedging instrument used in the strategy. This is why volatility options are proposed to hedge volatility risk (Whaley, 1993). Whether volatility options are more effective hedging instruments than plain vanilla options is documented by some recent research (Psychoyios and Skiadopoulos, 2006) and is beyond the scope of this paper. The use of ATM options as the hedging instruments in this paper ensures relatively stable results. The relative hedging performance of different models indicates a similar pattern as observed previously, that is, that the dollar hedging errors and MADs again demonstrate that the SV model outperforms the BS model in hedging short-term and medium-term barrier options and long-term ITM barrier options. Adding the jump feature to the model leads to poor hedging performance in terms of MADs.

To examine the effect of the barrier level on hedging effectiveness under this strategy, we consider ITM and ATM up-and-out call options with lowest, lower, higher, and highest barrier levels, as in the case of the minimum variance hedging strategy. The results are presented in Table IV. These results again confirm that hedging is most risky for options with the lowest barrier level.

The hedging results for call-on-call options are reported in Table VII. The results are also similar to those in the case of minimum variance hedging strategy. The SV and SVJ models perform better than the BS model and the JD model under this hedging strategy, based on the average dollar errors. This reconfirms our conclusion in the case of barrier options that the most

Table VII. Delta-Vega Hedging Errors for Compound Options

This table reports the delta-vega hedging errors for each model and for each category of call-on-call options. The underlying call option is a one-month at-the-money currency option. An option is a short-term option if its maturity is one month, a medium-term option if three months, and a long-term option if one year. BS, JD, SV, and SVJ stand for the BS model, the jump diffusion model, the stochastic volatility model, and the stochastic volatility and jump diffusion model, respectively. The strikes of the call-on-call options are set equal to 0.5, 1.0, 1.5, 2.0, and 2.5 times the one-month at-the-money option price, respectively.

Maturity	Moneyness (S/X)	Dollar Errors (%)				Mean Absolute Deviations (%)			
		BS	JD	SV	SVJ	BS	JD	SV	SVJ
Short term	0.5	-1.75	-1.92	-2.39	-2.82	3.30	3.27	4.94	4.09
	1.0	-2.18	-2.57	-2.39	-3.21	4.65	4.55	5.85	5.77
	1.5	-2.51	-2.96	-2.27	-2.95	6.28	6.10	7.22	7.38
	2.0	-2.74	-3.04	-2.13	-1.91	8.30	8.21	9.01	8.82
	2.5	-2.88	-2.76	-1.95	-0.04	10.94	10.42	11.13	9.92
Medium term	0.5	-0.53	-0.50	0.02	0.51	2.51	2.41	3.91	2.41
	1.0	-0.57	-0.52	0.11	0.64	2.98	2.90	4.34	2.73
	1.5	-0.58	-0.57	0.17	0.76	3.47	3.45	4.82	3.01
	2.0	-0.59	-0.62	0.23	0.90	3.97	4.05	5.39	3.36
	2.5	-0.65	-0.71	0.30	1.06	4.53	4.72	6.02	3.71
Long term	0.5	-0.41	-0.46	0.03	-0.05	1.46	1.35	2.45	1.40
	1.0	-0.40	-0.45	0.04	-0.05	1.58	1.46	2.69	1.50
	1.5	-0.38	-0.44	0.07	-0.04	1.71	1.56	2.93	1.60
	2.0	-0.36	-0.42	0.10	-0.03	1.86	1.67	3.18	1.72
	2.5	-0.33	-0.41	0.13	-0.02	2.03	1.79	3.44	1.85

important feature in improving the performance of the BS model is to build the stochastic volatility feature into the model. The hedging effectiveness of each model increases as the maturities of the compound options increase. Overall, regardless of the strategies adopted, the performance patterns of hedging compound options are similar to those of hedging plain vanilla options.

V. Conclusions

In this paper, we assess the relative performance of the BS model, the JD model, the SV model, and the SVJ model using a different method from the commonly used out-of-sample testing approach in the literature. The models are calibrated to match the market prices of currency options and are recalibrated whenever the hedging portfolios are rebalanced. More importantly, they are evaluated based on the effectiveness of hedging exotic options, such as up-and-out calls and call-on-call options. This methodology is practical and relevant, as option-pricing models are mainly used to price/hedge exotic or illiquid options. Moreover, exotic options are more sensitive to model risk because exotic options have much more complex payoff structures than European options. This paper also investigates how the exotic feature in the options considered may affect model performance.

Our results demonstrate that the performance of the alternative models relative to the BS model depends on how “exotic” the hedged option is. Barrier options are more path dependent than compound options; therefore, the model’s relative performance of hedging barrier options is quite different from that of hedging compound options. The SV model outperforms the BS model in hedging most types of up-and-out call options, while the JD and SVJ models generally demonstrate poorer performance in hedging barrier options. For hedging call-on-call options, the alternative models are usually better. The results underlie the importance of incorporating the stochastic volatility feature into the model framework. While it is documented in the literature that the hedging of vanilla options is relatively insensitive to model misspecifications as long as the hedging portfolio is rebalanced frequently, our results imply that choosing a suitable model is particularly important for hedging different types of exotic options.

In addition, for any given model, the hedging performance deteriorates as the exotic feature in the option becomes more important. For hedging up-and-out calls, both the barrier level and the maturity have a great impact on the hedging effectiveness. It is most risky to hedge up-and-out calls with a very low barrier level, and the hedging errors increase with the maturities for ITM up-and-out options. However, the maturity effect of hedging call-on-calls is similar to that of hedging plain vanilla options.

The option-pricing models considered in this paper cannot capture the unique feature of the stochastic skewness observed in the currency options market. It is worth exploring whether models that generate stochastic skewness (Carr and Wu, 2007) can further improve the hedging effectiveness in future research. Branger and Schlag (2008) investigate whether tests based on hedging errors can identify volatility risk premia, and they highlight the problems of such tests caused by the discrete adjustment to the hedge and model misspecification. It is recognized that volatility risk and jump risk are both present in addition to underlying price risk in the options market. If these risk factors are priced, what are the associated risk premium components in the hedging errors identified in this paper and how does the discretization and model error affect these components, thereby affecting our findings? These issues are interesting and left for future research. ■

Appendix. The Characteristic Functions in Equation (6)

The characteristic functions f_j ($j = 1, 2$) in Equation (6), which are derived by Bakshi, Cao, and Chen (1997), are given by

$$\begin{aligned}
 f_1 = \exp \Bigg\{ & i\phi((r-q)(T-t) + \ln S) - \frac{k\theta}{\sigma^2} \left[2 \ln \left(1 - \frac{(\xi_1 - k + (1+i\phi)\rho\sigma)(1-e^{-\xi_1(T-t)})}{2\xi_1} \right) \right] \\
 & - \frac{k\theta}{\sigma^2} (\xi_1 - k + (1+i\phi)\rho\sigma)(T-t) + \lambda(1+\mu) \left((1+\mu)^{i\phi} e^{0.5i\phi(1+i\phi)\delta^2} - 1 \right) (T-t) \\
 & - \lambda i\phi\mu(T-t) + \frac{i\phi(i\phi+1)(1-e^{-\xi_1(T-t)})}{2\xi_1 - (\xi_1 - k + (1+i\phi)\rho\sigma)(1-e^{-\xi_1(T-t)})} v \Bigg\}
 \end{aligned}$$

and

$$f_2 = \exp \left\{ i\phi((r - q)(T - t) + \ln S) - \frac{k\theta}{\sigma^2} \left[2 \ln \left(1 - \frac{(\xi_2 - k + i\phi\rho\sigma)(1 - e^{-\xi_2(T-t)})}{2\xi_2} \right) \right] \right. \\ \left. - \frac{k\theta}{\sigma^2} (\xi_2 - k + i\phi\rho\sigma)(T - t) + \lambda \left((1 + \mu)^{i\phi} e^{0.5i\phi(i\phi - 1)\delta^2} - 1 \right) (T - t) \right. \\ \left. - \lambda i\phi\mu(T - t) + \frac{i\phi(i\phi - 1)(1 - e^{-\xi_2(T-t)})}{2\xi_2 - (\xi_2 - k + i\phi\rho\sigma)(1 - e^{-\xi_2(T-t)})} v \right\},$$

with

$$\xi_1 = \sqrt{(k - (1 + i\phi)\rho\sigma)^2 - i\phi(i\phi + 1)\sigma^2}, \\ \xi_2 = \sqrt{(k - i\phi\rho\sigma)^2 - i\phi(i\phi - 1)\sigma^2}.$$

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